IFRS 9 Modelling Training

Pre-Read Material

Building a better working world

Agenda

1 Introduction

2 Staging

3 Lifetime Assessment

4 Low Default Portfolios

5 PD Modelling

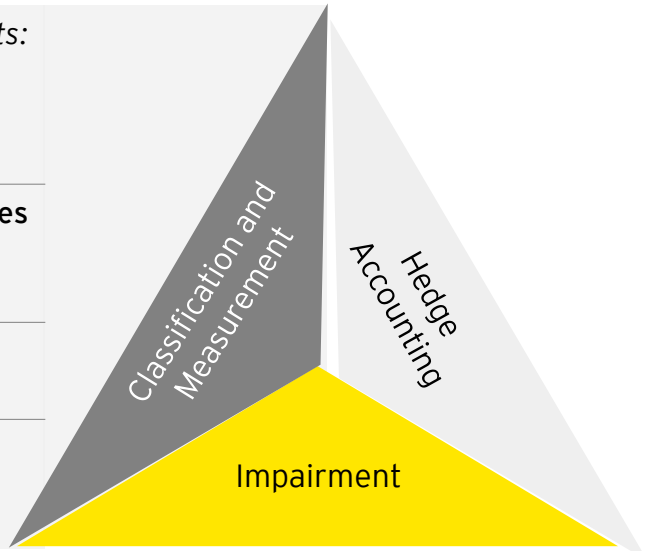
6 Forward Looking Adjustments

7 LGD Modelling

1 Introduction

Introduction
Background

- On **24 July 2014,** the IASB issued the final version of IFRS 9 *Financial Instruments* to replace IAS 39 *Financial Instruments: Recognition and Measurement*, one of the Standards inherited by the IASB when it began its work in 2001.
- IFRS 9 was developed following the financial crisis which highlighted the timelines of **recognition of expected credit losses (ECL)**, the **complexity of multiple impairment models** and **own credit** as areas in need of consideration.
- The package of improvements included in IFRS 9 are centred around three pillars as mentioned alongside.
- **The key focus of the training session is Impairment, and the expected modelling activities related to it.**



Introduction
Impairment

Need to address Impairment

- The incurred loss model under IAS **39 delayed the recognition of losses** until the loss event was triggered.
- As the financial crisis unfolded, it became clear that the incurred loss model gave room to **postpone losses**.
- Also, **complexity created due to use of multiple impairment** models was a concern.

	Benefits of a **Single Forward Looking Impairment Model**
1	The **forward looking impairment model** requires banks to recognize ECL **at all times** and update the amount **at each reporting date** to reflect the change in credit risk of financial instruments.
2	The model **eliminates the threshold for recognition of ECL**, so that there is no need for occurrence of a trigger event for the credit loss to be recognized.
3	The model requires a bank to base the measurement of its expected credit losses on **more timely and broadened information** which includes **forecasts** along with historical and current information.
4	The same impairment model is applied to **all financial instruments** that are subject to **impairment accounting**, thus, reducing unnecessary complexity.

Introduction

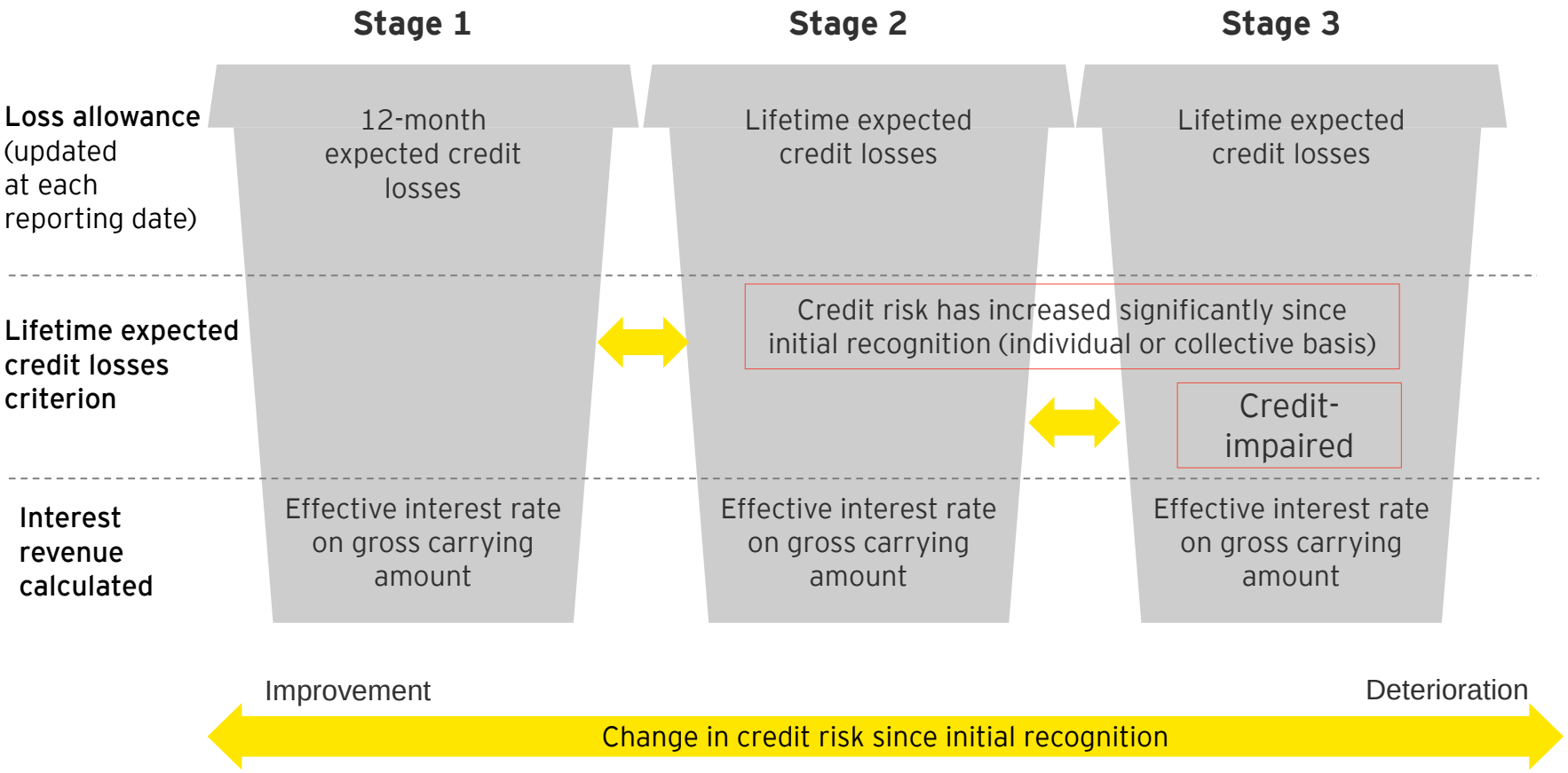
Impairment - IAS 39 vs. IFRS 9

Summary of Differences

		IAS 39	IFRS 9
1	Recognition of Credit Losses	▶ Recognition of credit losses is **delayed** until the **loss event was triggered**.	▶ The banks are expected to **make a provision** which serves as a proxy for the **initial expectations of credit losses**.
2	Use of Impairment Model	▶ The Incurred Loss Model is used which **delays the recognition of credit losses.**	▶ A forward looking ECL is required to be used under which the loss allowance is measured as either: **12-month ECL; or - lifetime ECL.**
3	Exposures to be included	▶ Loss allowances are only required to be measured for **impaired exposures.**	▶ Loss allowances are to be recorded for **all credit exposures** not measured at fair value through profit or loss.
4	Information Requirement	▶ The banks are expected to base its measurement of credit losses on **historical and current information.**	▶ The banks are required to use reasonable and **supportable forecast data** along with historical and current information.

Impairment Modelling

Application of General Approach



***Expected Credit Loss** = Exposure at Default (EAD) x Probability of Default (PD) x Loss Given Default (LGD)*

Expected Credit Loss calculation

Dual measurement approach

The model uses two approaches for measurement of expected credit loss:

12 month Expected Credit Loss (12M EL)	Cash shortfalls that will result if a default occurs within 12 months (or shorter period if the expected life is less than 12 months), weighted by probability of the default
Lifetime Expected Credit Loss (Lifetime EL)	Cash shortfalls that will result from default events occurring over the expected life (residual maturity) of financial instruments, weighted by probability of the default

$$ECL = \sum_{i=12M}^{LT} PD_i \times LGD_i \times EAD_i$$

Where:

- *12M are the estimates used for assets categorized into the Stage 1 of the General Approach; and*
- *LT are estimates used for assets categorized into Stages 2 and 3 of the General Approach*

The computed ECL amount is then discounted back to the reporting date using:

- *Effective Interest Rate (EIR) Undrawn loan commitments, Financial guarantees and all other assets; and*
- *Credit Adjusted EIR for assets that are impaired at purchase or origination*

Expected Credit Loss calculation

Definition of 12M and Lifetime ECLs

In practice

Expected Credit Losses should reflect the time value of money

- Expected cash flows should be discounted using the **effective interest rate** determined at initial recognition **or an approximation thereof**

From cash-flows to reporting date

From default to reporting date

- Not included in Basel
- 12 month or lifetime depending on the measurement stage

From cash-flows to default date

- Included in LGD
- But with varied discount rates – Cost of capital? Contractual rate? Other?

IFRS9 Introduction

Questions to Consider

- Is IFRS9 a regulation that impacts the accounting of a financial institution, or the capital of the institution?
- What is the essential purpose of the Classification and Measurement component of IFRS9?
- How often will an institution have to recalculate ECL and report it?

2 Staging

Staging

IFRS 9 Considerations

12m vs LECL

The usage of 12m or LECL for deciding provisions is driven entirely by staging, and hence, its importance cannot be understated

30 DPD

As per IFRS 9 requirements, a current delinquency of 30+ DPD is considered as a rebuttable presumption for movement to Stage 2, although it is considered a lagging and last resort indicator for significant deterioration in credit quality

Relativity

While the 30 DPD presumption is an absolute one, the general criteria for movement to Stage 2 is a 'significant deterioration in credit quality' since origination. As such, this is understood to imply a relative approach

Lifetime & Non PD

The relative increase in credit risk is meant to be measured over the entire lifetime rather than just 12 months. On a separate but related note, increase of (lifetime) PD is just one way of assessing this increase. Other relevant information should also be considered

General Viewpoints

- This area is considered the most judgmental and open-to-interpretation w.r.t. IFRS 9
- At the same time, its impact is the largest as it directly drives consideration of 12m or LECL
- Rebutting of the presumption will be unlikely and has not been seen so far. Recall rebuttability of 90 DPD definition of default
- There is a gray area in terms of how updated forecasts of macroeconomic data should be applied
- Internal team has not yet finalized a staging criteria, although it will be relative, based on a Lifetime PD at Reporting Date vs Lifetime PD at Origination multiple

Significant increase in credit risk

- The assessment of Significant Increase in Credit Risk is key in establishing the point of switching between measurement of 12M EL and Lifetime EL allowance
- However, tracking and determining whether there has been significant increases in credit risk of an exposures since initial recognition is also one of the major challenges in implementation of the ECL Model

Assessment

IFRS 9 states that credit risk analysis is a multi-factor and holistic analysis. Whether a specific factor is relevant, and its weight compared with other factors, will depend on:

- the type of financial instrument;
- Type of customer

Basis for Individual or collective basis of assessment

- A significant increase in **credit risk may not be evident on an individual instrument basis** before the financial instrument becomes past due
- In these cases, an assessment of whether there has been a significant increase in credit risk on an individual basis **would not faithfully represent** changes in credit risk
- So if more forward looking information is available on a collective basis, the entity should make the assessment on a collective basis

Significant increase in credit risk

Payments more than 30 DPD

- IFRS 9 contains a rebuttable presumption that the condition for recognising Lifetime EL is met when payments are more than **30 DPD**
- This presumption is **not an absolute indicator**, but should be considered as the **latest point** at which Lifetime EL should be recognised as delinquency is a lagging indicator,
- The presumption can be rebutted only if an entity has **reasonable and supportable information**
- RBI WG takes note that in most of Corporate/SME portfolios 30 DPD is a common phenomenon.

Exception for assets with low credit risk

- An entity may assume that the criterion for recognising Lifetime EL is not met if the credit risk on the financial instrument is low at the reporting date.
- IFRS 9 states that for a financial instrument, the credit risk is low if:
  - the borrower has a **strong capacity to meet its contractual cash flow obligations** in the near term; and
  - adverse changes in economic and business conditions in the longer term **may, but will not necessarily,** reduce the borrower's ability to fulfil its obligations
- In Indian context following criteria can be considered:
  - Central govt. guaranteed loans
  - Loans above a particular external/internal credit rating grade, showing **highest stability** and default rate **way below risk appetite** of the bank.

Staging

Questions to Consider

- Is an exposure that is in the Current DPD bucket always going to be in Stage 1?
- Can exposures move back and forth between the three stages set out in IFRS9?
- When can the 30 DPD backstop criterion be rebutted? What kind of analysis can help support the rebuttal?

3 Life time assessment

Life time assessment

Revolving Facilities

Problem statement

- Defining maturity can be extremely challenging, it is subject to interpretation of the standard and depends on product and behavioural characteristics
- There is still a lot of uncertainty around the lifetime calculations for revolving and renewable facilities

Challenges

1. How is lifetime defined for credit cards or revolving loans with no contractual maturity?
2. Is a standard value – e.g. 12 months – acceptable in Stage 1? And Stage 2, or is lifetime shorter?
3. How are cures incorporated?
4. Interactions between Stage 2 triggers / Risk management actions / Horizon of risk

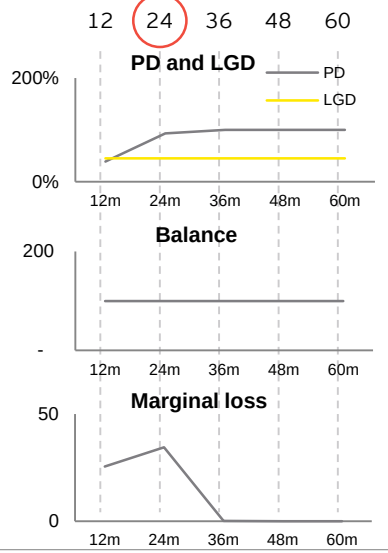
Practical example

Loan type	Credit card
Balance	Rs.1000
Limit	Rs.2000
12m PD	38%
LGD	45%
Stage	Stage 2
Maturity	No contractual mat

For an account in a revolving financial product which is in stage 2, the lifetime can be determined by:

1. Looking at historical, experienced lifetime for accounts that have shown deteriorating credit quality in similar products – in this case we assume a 'at risk' of 24 months
2. Assume 12 months lifetime

?



The Lifetime expected loss is the sum of the marginal losses throughout the expected lifetime. Assume that the lifetime for this account is set to 24 based on historical experience.
Marginal Loss Year 12m = 26
Marginal Loss Year 24m = 35
Total loss = 61

Life time assessment

Term Loans

Problem statement

- Whilst most banks will be using the contractual date when calculating lifetime EL for other products, there are some instances where it would be beneficial to use the behavioural lifetime. Due to typically long contractual lifetime, mortgages are the product where the gap between both dates is most significant – there are different views on how to perform the calculations: cap maturity or incorporate pre-payments on EAD profile?

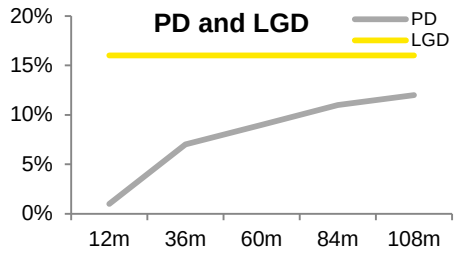
Challenges

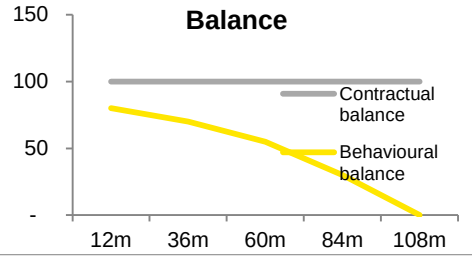
1. How is the contractual lifetime defined when the market policy is to renew loans at product maturity date?
2. How can lifetime be estimated on significantly deteriorated accounts?
3. Do we truncate maturity, or do we use the probability of re-structuring/pre-payment and work with contractual maturity?

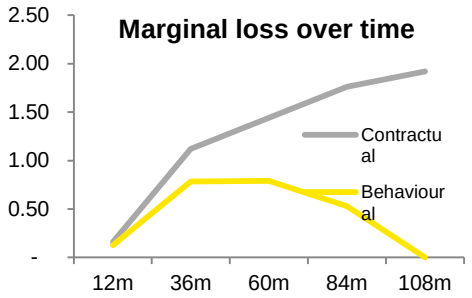
Practical example

If it can be shown that the behavioural lifetime is shorter than the contractual lifetime the provision held for those portfolios will decrease.

If a behavioural pattern of prepayments can be modelled that will further improve the accuracy of the impairments







Lifetime Assessment

Questions to Consider

- Are there any portfolios for which separate pre-payment models may be required / justified?
- What kind of business knowledge / product understanding can be leveraged to determine behavioural lifetimes for revolving products?

4 Low Default

Types of low default, low loss and non-modelled portfolios

These assets has never experienced any default or really few defaults over the last decade therefore traditional statistical methodology such as logistic regression is not suitable. The current regulatory approach for these assets is quite conservative and not very advanced. Most of those assets are reported under the Standardised or F-IRB approaches.

1 Loans and advances to banks	• Assets that behave as collateral call accounts related to Credit Support Annex (CSA) agreements or operational accounts for clearing and other purposes • Characterised by very short maturities and fluctuating balances
2 Primary liquidity bonds and Sovereign / Quasi-Sovereign	• Assets typically held to collect contractual cash flows and improve running or redemption yields by periodically moving from one holding to another • For example, UK gilts, supernatural debt securities and non-domestic government bonds
3 Structured Finance	• Refers to pools of illiquid assets which can not be sold individually; for e.g. ABS, RMBS, CLO and Covered Bonds • Overall risk of the pool is 'diversified' among different investors through securitization, in order to help banks remove risky assets from the balance sheet.
4 Government supported assets	• Assets are supported by local government initiatives such as Social Housing; therefore they are low risk and not likely to default • They may witness higher volatilities depending on policy uncertainties or changes in government

Low Default Portfolios – challenges

*A portfolio can be classified as a Low Default Portfolio ("LDP") when a bank's internal data systems include fewer loss events, which presents challenges for risk quantification and validation.**

IFRS 9 Challenges

1 **Conservatism estimates**	• Common approaches for LDP are conservative by nature / design • Producing a best estimate of default will require stripping away conservative overlays, which may be expert judgement driven and can be challenged
2 **Forward looking**	• Embedding cyclical, forward-looking predictions in producing Expected Credit Loss estimates are a major analytical challenge • Macro-economic factor correlation to observed defaults may be weak • Not enough historical data differentiation available to make a probability-weighted best estimate outcome

3 **Model risk**	• Different methodologies and assumptions can lead to significant variances in the PD estimates
	• Data availability constraints can limit:
	  • Discriminatory power of models
	  • Distribution of observations (Poisson) - may not be tractable
	  • Difficult to back-test PD estimates

4 **Data availability**	• Recovery data more sparse than default data, and common methodology mainly deal with PD • Qualitative assumptions that can be challenged by auditors and hard to justify • Use of external data may not be full representative of portfolio and may be challenged by auditors

* Source: Basel Committee Newsletter No. 6 (September 2005)

Low Default Portfolio – potential IFRS 9 approaches

Strong model governance across Risk and finance will be required to ensure the bank is comfortable with the approach taken and its potential limitations

			Challenge potentially addressed			
Example LDPs	**Potential approaches**	**IFRS 9 application**	**1**	**2**	**3**	**4**
Sovereign	External ratings	• Use of implied PDs from external rating agencies • An average rating from a number of agencies could be used	✓	✓	✓	✓
Corporates	Reduced form approach (Jarrow-Turnbull)	• Sufficiently forward-looking, as the yield curve reflects macro-economic expectations • Use of external market observable data points (credit spreads)	✓	✓	✓	✓
Corporates/FIs	Structural approach (Merton Model)	• Presents a “best estimate” PD based on all present and future information available on firm’s assets and liabilities • Presents a relation between unobservable variables (future market values of assets) and observable variables – makes the approach sufficiently forward-looking • Recovery rates are based on a forward-looking expectation of default	✓	✓	✓	✓
Mortgages	Pluto-Tasche approach/BCR	• Confidence level can be suited to “best estimate” point-in-time instead of a conservative view	✓	✓	✓	✓

1 - Conservatism estimates, 2 - Forward looking, 3 - Model risk, 4 - Data availability

✓ = Fully addressed ✓ = Partially addressed

Non-modelled and Standardised portfolios – potential solutions

Key Questions	Potential solutions
• Data is available, but planned model development is beyond IFRS 9 timelines	• Accelerate model development timeline to meet IFRS 9 needs
• Lack of internal historical experience (e.g., new product)	• If risk characteristics are similar, use other product data as a proxy • Leverage external data if available
• Lack of observations for the segment	• Re assess segment hierarchy to increase data observations (eg, reduce segmentation granularity) • Leverage external data if available
• Standardised portfolios	• Use non regulatory approved models if available • Leverage existing IRB models • Leverage external data if available
• Immaterial portfolio or closed book	• Define materiality framework which also considered risk • Assess the size of closed portfolio at time of “go live”

Specialized Lending - Definition

SL exposures possess all of the following characteristics, either in legal form or economic substance:

- The exposure is typically to an entity [often a special purpose entity (SPE)] that is created specifically to finance and/or operate physical assets;
- The borrowing entity has little or no other material assets or activities, and therefore little or no independent capacity to repay the obligation, apart from the income that it receives from the asset(s) being financed;
- The terms of the obligation give the lender a substantial degree of control over the asset(s) and the income that it generates (or in other words the subject loan represents a significant liability in the borrower's capital structure); and
- As a result of the preceding factors, the primary source of repayment of the obligation is the income generated by the asset(s), rather than the independent capacity of a broader commercial enterprise.

Types of Specialized Lending

1. Project Finance
2. Object Finance
3. Commodity Finance
4. Commercial Real Estate

Supervisory Slotting Criteria

As per the Basel II guidelines, banks that do not meet the requirements for the estimation of PD under the corporate IRB approach will be required to map their internal grades to five supervisory slotting categories, each of which is associated with a specific risk weight.

‘Supervisory slotting criteria’ are indicative risk drivers that the bank should consider while assessing the risk profile of the Specialised Lending transactions.

In developing the Specialised Lending rating models, adequate care should be taken to consider the supervisory slotting criteria covered under the Financial and Qualitative Risk categories. Basel II has broadly classified the criteria into 5 categories as mentioned in the table below.

Project Finance		**IPRE**	
Basel Criteria	**Coverage in Model**	**Basel Criteria**	**Coverage in Model**
Financial Strength	Financial Risk & Project Risk	Financial Strength	Financial Risk & Project Risk
Political & Legal Environment	Project Risk & Sponsor Risk	Asset characteristics	Property Risk
Transaction Characteristics	Project Risk	Strength of Sponsor / Developer	Sponsor / Developer Risk
Strength of Sponsor	Sponsor Risk	Security Package	Risk Mitigants & Financial Risk
Security Package	Risk Mitigants & Financial Risk		

Mapping of Rating Methodology to Supervisory Slotting Criteria

As per the Supervisory Slotting Criteria, if the bank does not have sufficient data for estimating PD, Unexpected Loss and Expected loss must be calculated based on risk weight associated with each supervisory category.

The bank can use supervisory risk weights for estimation of capital required for Expected and Unexpected losses. The supervisory categories broadly correspond to a range of external credit assessments as outlined below:

Supervisory Slotting Category	Strong	Good	Satisfactory	Weak	Default
Rating Grades	BBB- or better	BB+ or BB	BB-or B+	B to C-	N.A
Risk Weight for UL	70%	90%	115%	250%	0%
Risk weight for EL	5%	10%	35%	100%	625%

The rating grades for all of the above Specialized Lending models are mapped to the regulatory rating grades.

Risk Grades	Score range	Definition of borrower risk level	Mapping to Supervisory Grades
Above 80%	AAA	Very Low	Strong
Above 70% & up to 80 %	AA	Low	Good
Above 60% & up to 70 %	A	Moderately Low	Good
Above 55% & up to 60 %	BBB	Fair	Satisfactory
Above 50% & up to 55%	BB	Moderate	Satisfactory
Above 45 & up to 50%	B	Moderately High	Weak
40% and above & up to 45 %	C	High	Weak
Below 40%	D	Very High	Default

Illustrative

Low Default Portfolios

Questions to Consider

- Based on prior experience, does any on have any alternate suggestions in terms of methodology?

5 PD Modelling

Default

IFRS 9 Considerations

Definition

The definition for identification of default continues to be 90 Days Past Due; it is rebuttable, although most banks in the market are not choosing to do so, unless they have already done it as part of their Basel exercise

Horizon

A 12 month PD, as required for Basel, continues to be a requirement for IFRS 9; additionally, and more importantly, a lifetime PD is also required to be calculated and kicks in depending on the stage of a loan

Lifetime

Lifetime is considered as remaining contractual term for amortizing products; for revolving facilities and facilities with off-balance sheet components, behavioural lifetime needs to be derived rather than just considering remaining contractual term

Introduction to PD Modeling for IFRS 9 (2/2)

- Designing the PD methodology for IFRS implies defining the models or methodologies for the following:

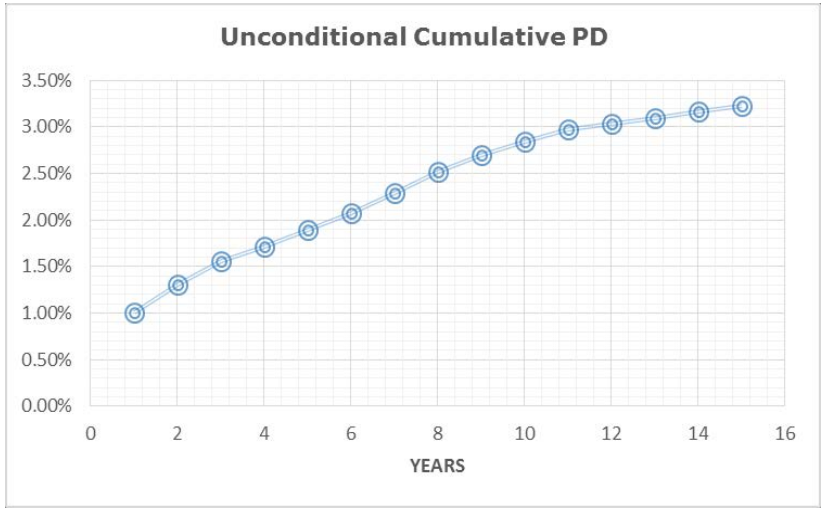
1 Developing Unconditional PD Curves

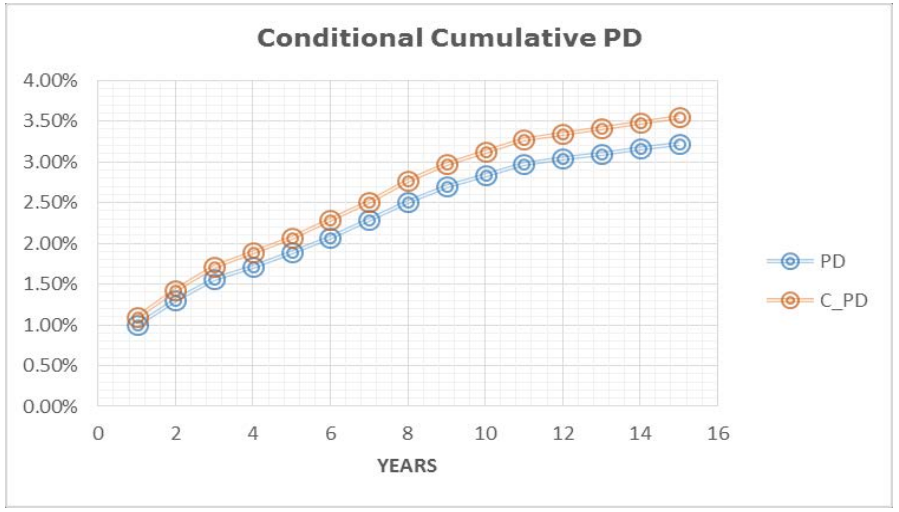
- IFRS 9 requires a lifetime PD. This means that there should be some mechanism of projecting the Unconditional PD, Marginal or Cumulative, on a forward looking basis.
- By "Unconditional PD" we mean that this does not take into account the impact of forward looking macro-economic scenarios and is independent of future macro forecasts.

2 Developing Methodology to Condition the PD based on Macro Factors

- The Unconditional PD must then be conditioned to develop Conditional Marginal PD.
- By "conditional" what we mean is that the PD at a certain forward point in time is conditional or dependent on the forecast of the economic conditions at that point in time
- This conditioning would be based on forecast of macro-economic factors and other relevant forward looking information.

Illustration of Unconditional and Conditional PD and their linkage





Macro- Economic Forecast Oil Price, GDP, Growth etc.

+

Link Function to link unconditional PD and Macro- Economic Factor

Retail PD modeling

1. Possible Options for Unconditional PD Curves

- There are several feasible options for developing Unconditional PD curves with varying levels of complexity and sophistication. We have detailed them below:
  - Constant Marginal PD rate
  - Using Flow Rate Matrix
  - Vintage Curve Analysis
  - Scorecards
- We then compare these options to decide on the preferred approach

Constant Marginal PD Rate	▸ This is the simplest method and assumes a constant Marginal PD rate. The Marginal PD rate can be a 12-Month PD derived using historical default data. ▸ This method is very easy to implement but is simplistic in its assumption of constant Marginal PD rate ▸ $UnconditionalCumulativePD(YearN) = 1 - (1 - MarginalUnconditionalPD_{Yr1})^N$
Using Flow Rate Matrix	▸ An alternative is to use the Flow Rate Matrix. ▸ Typically Flow Rate Matrices are monthly in nature and there to develop the Cumulative PD for a certain horizon, say N months, we need to multiply N matrices. ▸ This is a conceptually simple method and , given that Flow Rate Matrices are commonly used in retail portfolio management, is also easy to implement. ▸ However, this method has some drawbacks: ▸ It becomes difficult to implement as N increases. ▸ In addition, we need to make the assumption that the Flow Rate Matrix is constant. ▸ Also, Flow Rate Matrices are based on delinquency buckets only and cannot easily take into account other risk factors

1. Possible Options for Unconditional PD Curves

Vintage Curves	▸ Retail portfolios typically demonstrate a strong Vintage and/or Month on Book (MOB) effect. This means that the default rates of the portfolio have a strong linkage to the lifecycle stage of the portfolio. ▸ Therefore when we develop the PD curves for the retail portfolio, the impact of the Vintage on the Shape of the Curve must be kept in view. ▸ Vintage curves have been traditionally used in classical retail portfolio management. These curve depict the movement in Cumulative Default Rates vs MOB ▸ Vintage curves therefore are the most robust method for deriving Unconditional PD Curves and would be the preferred methodology
Scorecards	▸ An approach that EY has experience in, and is increasingly seen as the preferred methodology adopted across a few regions ▸ Involves building standard A and B scorecards; lifetime PD estimates are then derived by calibrating model output (12m PD) to a range of lifetime values (1m to 60m, for e.g.)

Decisioning approach

- Given the 3 options defined above with varying levels of complexity and sophistication the choice can be based on following:
  - Vintage Curves will be the most preferred option. However, this would be subject to robustness, adequacy and quality of data.
  - Flow Rate based PD curve can be used for accounts in higher delinquency buckets like 30+ and 60+ since these have a steep role into default and near term risk dominates lifecycle stage risk. However, this would not be a preferred option fo r Current and 1-30 DPD buckets
  - Constant Marginal PD rate assumptions will be used in a scenario where we are not able to develop robust Vintage Curves

Developing Vintage Curves

A Range of Options to Choose From

Technique	Advantages	Disadvantages
Loss Rate Approach	▶ Implicitly incorporates LGD; no need to model LGD separately ▶ Straightforward and simple to implement	▶ Tends to overestimate ECL ▶ Cannot adjust for dynamic portfolio origination and management strategies
Constant Marginal PD	▶ Most intuitive and easy-to-understand approach ▶ Straightforward and simple to implement	▶ Key assumption of constant marginal PD too simplistic and not realistic
Survival – Non Parametric	▶ Intuitive and easy-to-understand approach ▶ No room for mistakes / incorrect interpretations / technical nitty-gritties ▶ Straightforward and simple to implement	▶ Cannot incorporate effects due to static factors or time varying effects
Survival – Semi Parametric	▶ Incorporates best of Non Parametric and Parametric; highly robust ▶ Can account for differentiation due to static factors as well as time varying effects	▶ Has quite a few assumptions that need to be tested ▶ Interpretation in some cases can be unclear ▶ Implementation can be challenging
Survival – Parametric	▶ Good at smoothing data inconsistencies to get a well behaved PD curves ▶ Can account for differentiation due to static factors	▶ Cannot account for differentiation due to time varying effects without introducing significant complexity

Increasing order of sophistication

EY

Loss Rate Approach

Simplistic

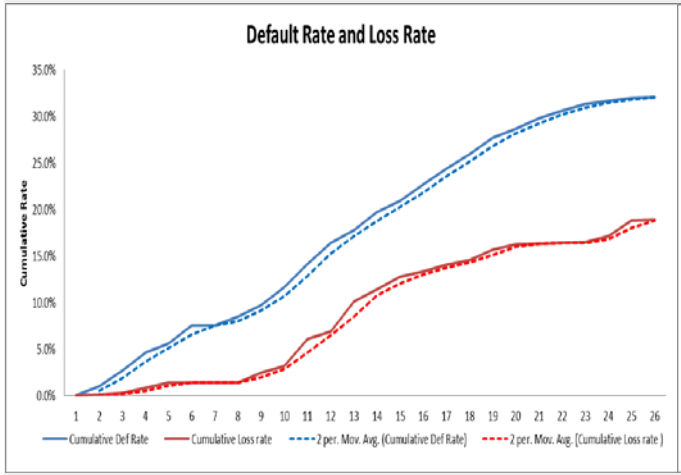
Highlights

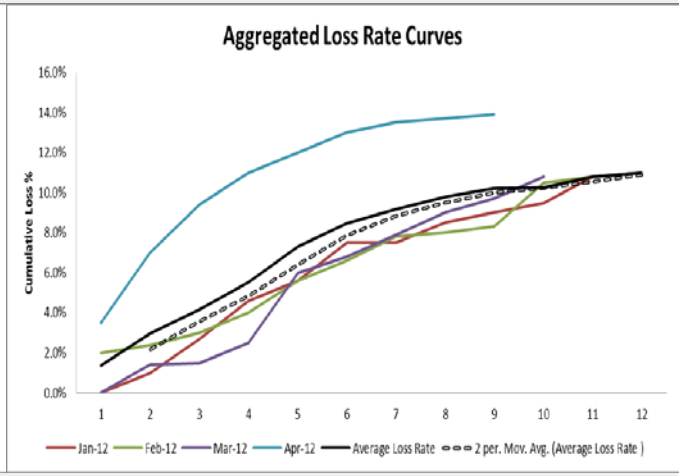
- Choose a particular snapshot month and all loans originating in said snapshot. Chosen snapshot should be at least far as the typical tenure of a loan for robust estimates
- Track accounts for all subsequent months. Calculate cumulative and marginal (single time period) default and loss rates keeping the base fixed as at start of snapshot month
- Repeat this exercise for all available snapshots to ensure TTC estimates of default and loss rates
- 12m and lifetime default and loss rates can be aggregated across all snapshots using a suitable logic to obtain aggregated rates which can then be used for ECL computation

MOB	1	2	3	12	13	14	26
Accounts At Obs	1000	1000	1000	1000	1000	1000	1000
Balances At Obs	Rs. 20,30,000	Rs. 20,30,000	Rs. 20,30,000	Rs. 20,30,000	Rs. 20,30,000	Rs. 20,30,000	Rs. 20,30,000
#DefaultAccounts	0	10	17	23	14	19	1
Default balances (EAD)	Rs. 0.00	Rs. 1,500.00	Rs. 3,000.00	Rs. 16,000.00	Rs. 67,000.00	Rs. 25,000.00	Rs. 1,056.00
#CummDefs	0	10	27	23	37	56	321
#CummDefEAD	Rs. 0.00	Rs. 1,500.00	Rs. 4,500.00	Rs. 16,000.00	Rs. 83,000.00	Rs. 1,08,000.00	Rs. 3,83,123
Marginal DefRate	0.00%	1.00%	1.70%	2.30%	1.40%	1.90%	0.10%
Marginal LossRate	0.00%	0.07%	0.15%	0.79%	3.30%	1.23%	0.05%
Cumulative Def Rate	0.0%	1.0%	2.7%	2.3%	3.7%	5.6%	32.1%
Cumulative Loss rate	0.00%	0.07%	0.22%	0.79%	4.09%	5.32%	18.87%

12 Month LR and Default rate

Life time Month LR and Default rate





Constant Marginal PD Approach

Simplest and Most Intuitive Approach

Highlights

- Assumes a constant marginal PD for the lifetime; the marginal PD is calculated from TTC 12m PDs spread across relevant dimensions, including delinquency
- This constant marginal PD is then applied on a conditional basis to obtain lifetime ECL estimates
- This method is best suited for obtaining quick estimates of ECL as and when needed. For e.g., the first round of Quantitative Impact Assessment results

	Current	X DPD	30 DPD	60 DPD
Personal Salary Transfer	2%	3.5%	8%	50%
Personal Non-Salary Transfer	3%	5.75%	12%	51%
Auto LTV > 80%	4%	7.5%	18%	59%
Auto LTV < 80%	2.5%	4.75%	11%	57%
Mortgage Tenure < 20Y	1.5%	2.8%	6%	52%
Mortgage Tenure > 20Y	2.75%	4%	10%	55%
Credit Card 6m Avg. Utilization < 65%	4.5%	6.5%	13%	58%
Credit Card 6m Avg. Utilization > 65%	5.5%	7%	15%	60%

$$PD_{in} = PD_i * (1 - PD_i)^{n-1}$$

Survival Analysis

Some Basic Concepts

- Survival analysis is a statistical method for data analysis where the outcome variable of interest is the time to the occurrence of an event, often referred to as a failure time, survival time, or event time.
- Suppose that T is the length of time before a facility defaults. The randomness of T can be described in three standard ways
  - Distribution function describes *F(t)* the probability the time to event (T) is smaller or equal compared to a fixed time (t) and is given as: $F(t) = P(T \leq t)$
  - From this, the survival function *S(t)* , the probability the time to event (T) is larger compared to a fixed time (t), can be derived as: $S(t) = 1 - F(t)$
  - The second way is the density function . This is the probability that the failure time occurs at exactly time t (out of all possible times) and is given as:

$$f(t) = \lim_{\Delta t \downarrow 0} \frac{Prob(t \leq T \leq t + \Delta t)}{\Delta t}$$

  - Hazard function. This is the probability that if a facility survives up till time t, it will experience the event in the next instant and is given by:

$$h(t) = \frac{f(t)}{1 - F(t)} = \frac{f(t)}{S(t)}$$

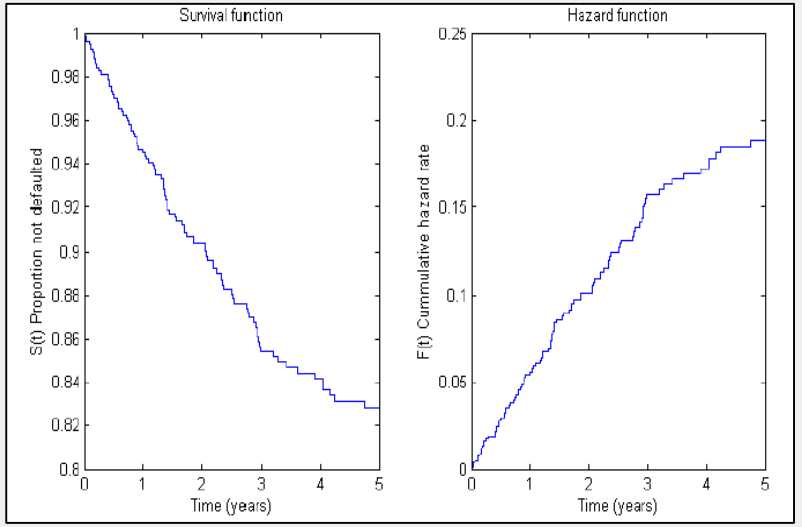
Survival Analysis

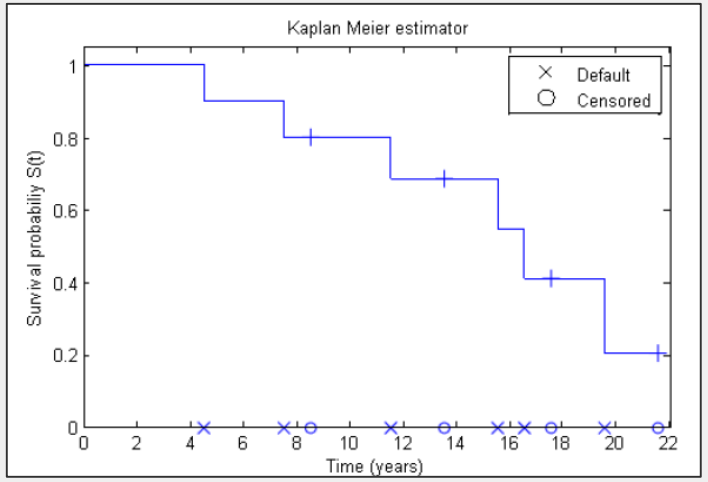
Non Parametric Approach (Kaplan-Meier Estimate / Curve)

Highlights

- Y axis of left figure represents proportion of population surviving (not defaulting); for right figure, it represents hazard rate (defaulting)
- X axis of both figures represents time scale (years on book in this case)
- For a given hazard rate, such as h (2) equalling 1%, the interpretation is that a customer having not defaulted for 2 years on book, has a 1% probability of defaulting in the next year
- In SAS, the LIFETEST or PHREG procedures can be used to obtain the Kaplan-Meier estimates / curves

$$\hat{H}(t) = \sum_{i:t_i<t} \frac{d_i}{n_i}$$



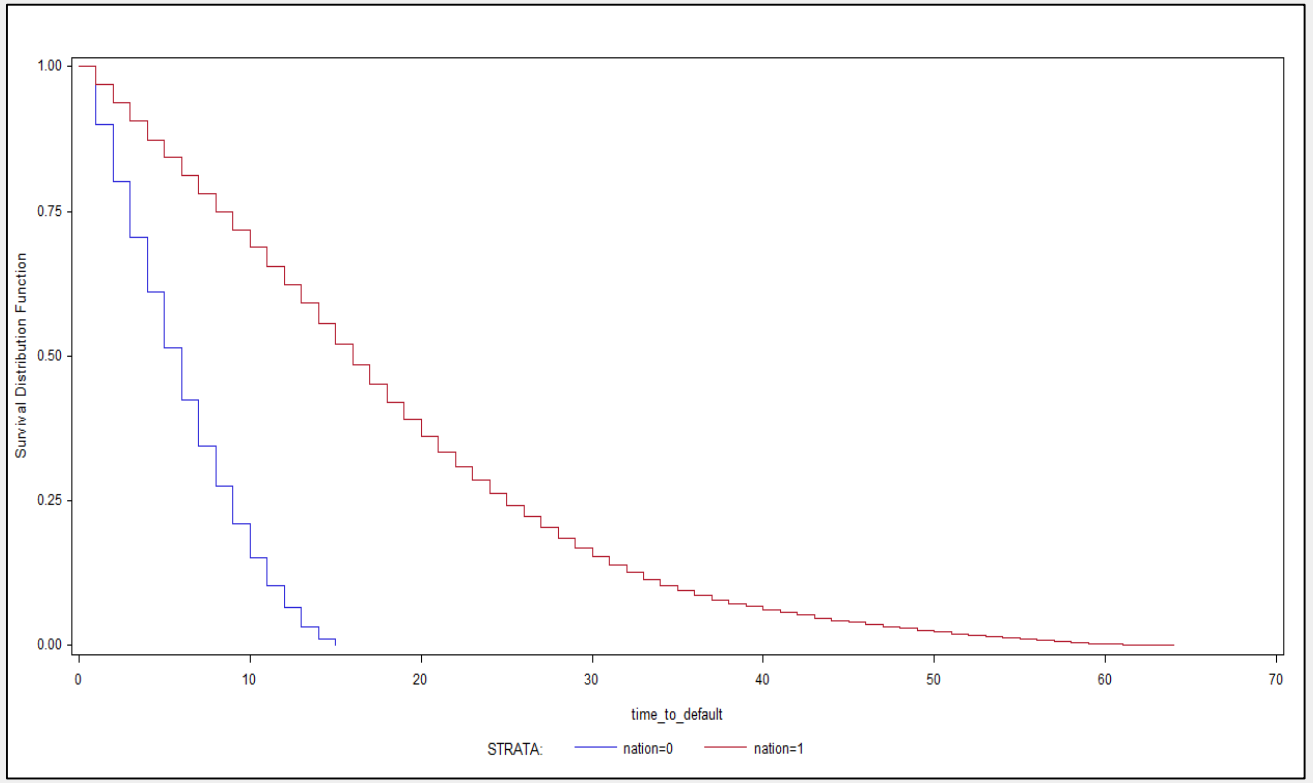


Survival Analysis

Semi Parametric Approach (Cox Proportional Hazards Regression)

Highlights

- Using the same hazard rate notation from before, the closed form for the Cox Proportional Hazards Regression is given by $h(t, X) = h_0(t) \exp(bX)$
- Here, $h_0(t)$ represents the baseline hazard rate function, which is non-parametric and derived from the Kaplan-Meier estimate
- The exponential term represents the effect of static and / or time varying effects. In the figure on the right, the static factor considered is Local / Expat tag, from which the different in riskiness is clear as one observes that by 10 MOB, only ~20% of the Expats have survived, whereas the same survival rate for Locals is ~70%. The beta coefficient associated with this tag variable will help bring out this differentiation
- In SAS, the PHREG procedure can be used to obtain the Cox Proportional Hazards Regression based models

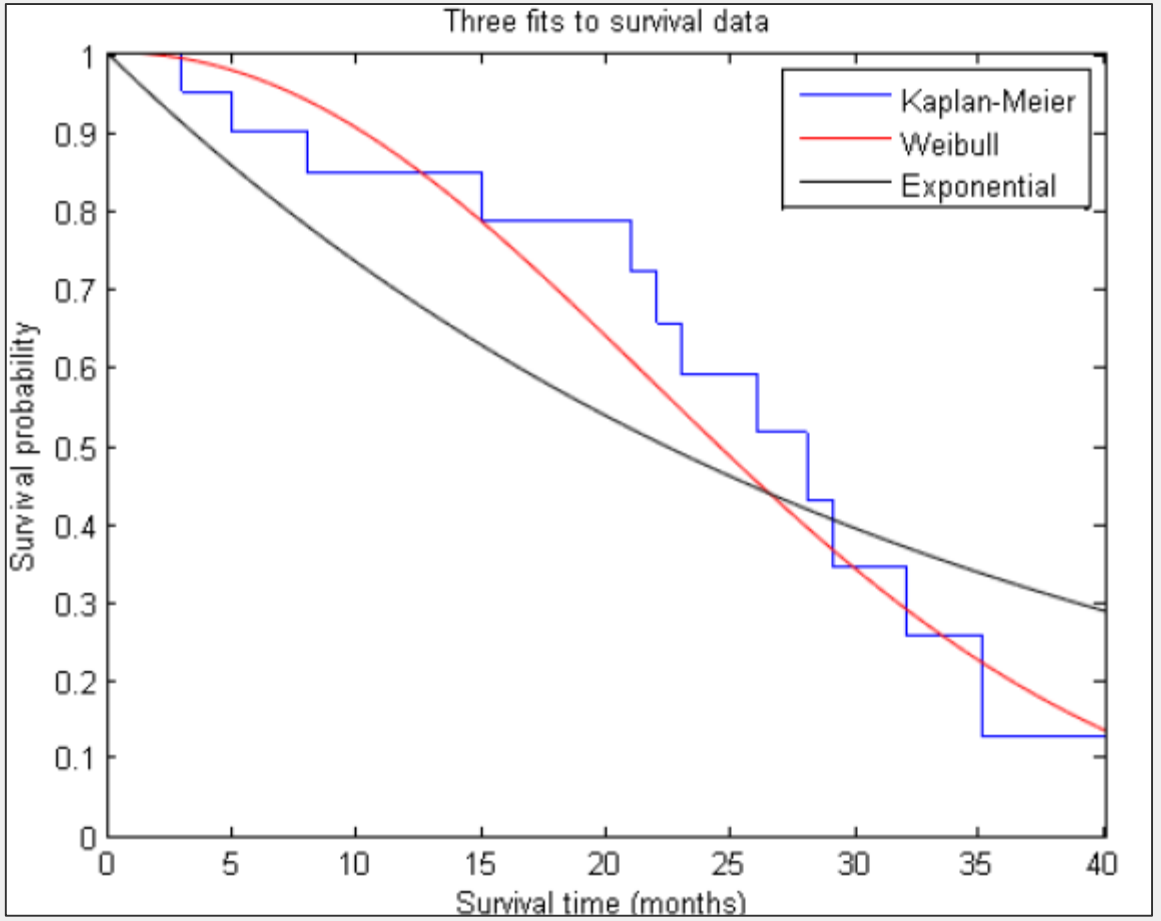


Survival Analysis

Parametric Approach (Distributional Fitting)

Highlights

- This class of survival models don't have any baseline hazard rates / function. Instead, an appropriate distribution is fitted to the empirical survival function
- The parameters of the fitted distribution are, in turn, dependent on a combination of static factors, which are finalized through a regression technique
- Some popular choices of distribution for fitting include Weibull Log Normal, Exponential, Gamma, Log Logistic, Normal, Logistic
- In SAS, the LIFEREG procedure can be used to fit different parametric distributions and obtain models



1.B. Adjusting Vintage Curves for Near Term Risk

- Vintage curves essentially reflect Long Term portfolio effects. Notwithstanding the Long Term effect of vintage or MOB on shape of the Unconditional PD curve, we also observe that delinquency information is usually a very strong near term indicator of default.
  - Variables like ever 30+ in last 6 months are used frequently in behaviuor scorecards. In addition, it is also seen that flow rates from 30+ or 60+ buckets into default are usually pretty steep.
  - IFRS 9 also has suggested the use of 30+ DPD as a rebuttable criteria for assessing Significant Increase in Credit Risk (SICR).
- One of the limitations of Cox PHR is that it cannot handle delinquency variables effectively. Therefore, the Vintage Curves need to be adjusted to reflect the effect of such immediate account level indicators.
- This would be done in a 2 Step Process:
  - Develop a Near Term Risk Scalar using delinquency based variables
  - Scale the Vintage Curve using the Near Term Risk Scalar
- This process is explained in the next few slides

Note: As an alternative to NTRS adjustment, we can define Vintage Curves for 30+ delinquency and use it. But this does not bring into play the MOB effects

1.B.(i) Developing a Near Term Risk Scalar

- A Near Team Risk Scalar (NTRS) is developed using Logistic Regression run on the portfolio against a set of Delinquency Related variables
  - Some examples of such variables (X's in the regression would be) Max DPD in last 6 months, Number of times 30+ DPD in last 12 months, Ever 60+ DPD in last 6 months etc.
  - The regression form would be as follows:
  - Log (PD/1-PD)= A + B1.X1+B2.X2+…..+Bn.Xn, where Xi indicates the delinquency variables
- The Logistic Regression would be calibrated against 12 months PD based on history of last 5 years
- The 12-Month PD for any account coming from the Logistic Regression, based on the delinquency status of the account, would be the NTRS for the account as shown below.
- **Go through the scorecard training deck for more details on how these models are developed**

NTRS = Exp(A + B1.X1+B2.X2+…..+Bn.Xn)/(1+Exp(A + B1.X1+B2.X2+…..+Bn.Xn))

Account x	**Set of Delinquency related** **Account X (X1,X2,…,Xn)**	**S= A + B1.X1+B2.X2+…..+Bn.Xn**	**NTRS = EXP (S)/ (1+EXP(S))**

1.B.(ii) Scale the Vintage Curve using NTRS

- Once the NTRS has been developed, the next step is to scale the Vintage Curve using the NTRS
- This is done as follows:
  - The Vintage Curve is normalized using the 12-Month PD. What this means is that every point on the Vintage Curve is divided by the 12-Month PD to bring it to a 12-Month PD equivalent.
  - In the next step the normalized 12-Month PD equivalent is multiplied by the NTRS to give the NTRS adjusted PD

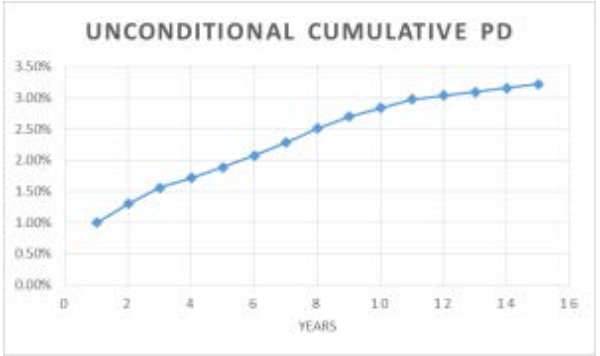
Month	PD	NTRS 12- Month Scale	2% NTRS Adjusted
6	1.00 %	77 %	1.54 %
12	1.30 %	100 %	2.00 %
18	1.56 %	120 %	2.40 %
24	1.72 %	132 %	2.64 %
30	1.89 %	145 %	2.90 %
36	2.08 %	160 %	3.19 %
42	2.28 %	176 %	3.51 %
48	2.51 %	193 %	3.87 %
54	2.70 %	208 %	4.16 %
60	2.84 %	218 %	4.36 %
66	2.98 %	229 %	4.58 %
72	3.04 %	234 %	4.67 %
78	3.10 %	238 %	4.77 %
84	3.16 %	243 %	4.86 %
90	3.22 %	248 %	4.96 %

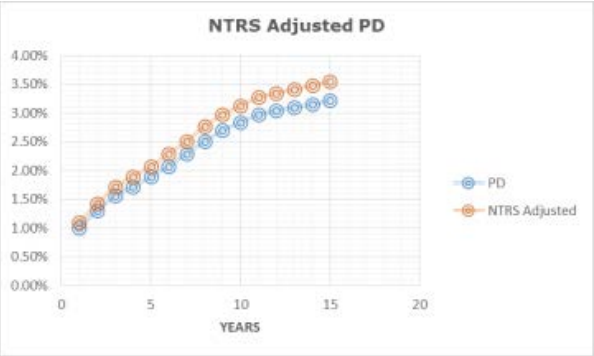
12-Month Scaled= 1.72%/1.30% = 132%

NTRS Adjusted = 132% * 2% (NTFS) = 2.64%

1. Bringing it all together for Unconditional PD Curves

- So to put everything together till this stage,
  - Every account can be mapped to a specific Vintage Curve. Given the MOB of the account we would know the remaining life PD curve for the account
  - Every account will also be mapped to a Near Term Risk Scalar based on its delinquency status
  - The above will be combined to give the NTRS adjusted unconditional PD curve for the account





Account X

Set of Delinquency related variables for Account X (X1, X2,..., Xn)

S= A + B1.X1+B2.X2+.....+Bn.Xn

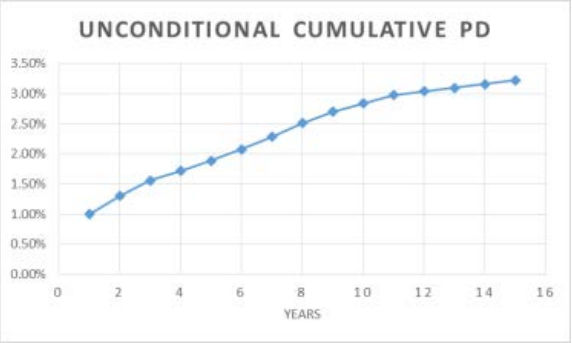
NTRS = EXP(S)/ 1+EXP(S)

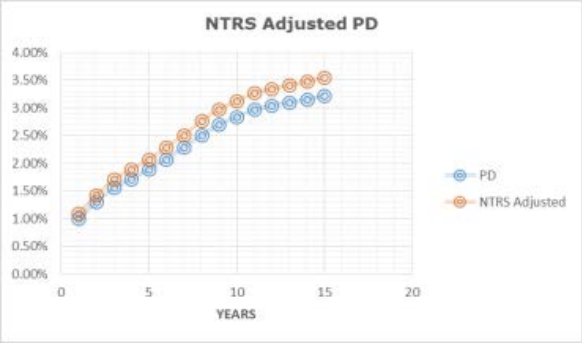
2.Conditioning PD based on macro factors

- In the previous section, we had derived the Unconditional PD curve. In this section we talk about how to condition the Unconditional PD curve based on macro-factors.
- The process of conditioning of PD curve based on macro factors is a 2 step process:
  - Determining which macro variables affect PD and their forward looking scenarios
  - Determining the Link Function between macro factor and PD
- The Link Function establishes the link between the Macro Factors and the Unconditional PD to give the Conditional PD
- There are two possible and commonly used options for Link Functions:
  - Vasicek Framework
  - Regression Based Framework
- Vasicek Framework is more suited for Non-Retail and not commonly used in Retail. In addition, the amount of internal default data available is more Retail as compared to Non-Retail. Therefore we will go for a Regression based Framework

3. Summary of Retail PD modeling

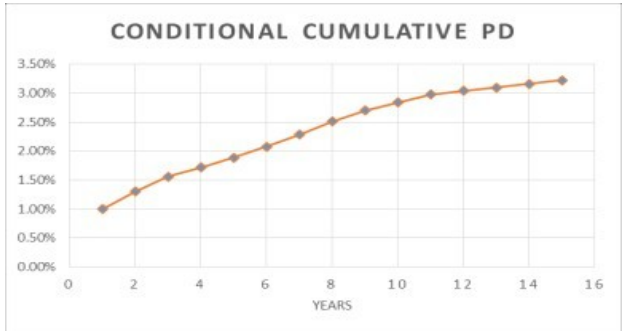
1





4

- $ConditionalPD_{yr3} = UnconditionalPD_{yr3} + \frac{\Delta PD}{\Delta GDPGrowth} * (GDP_{yr3} - TTCPD)$



3

Account or
Risk Segment Level

2

Set of Delinquency related variables for Account X (X1, X2,..., Xn)

NTRS =
EXP(S)/ 1+EXP(S)

S= A + B1.X1+B2.X2+.....+Bn.Xn

Non- Retail PD modeling

1. Possible Options for Unconditional PD Curves

- There are several feasible options for developing Unconditional PD curves with varying levels of complexity and sophistication.
  - No Rating Transition (Constant Marginal PD rate)
  - Assumes Rating Transition Matrices

No Rating Transition (Constant Marginal PD Rate)	▸ This is a straightforward method and assumes a that given a rating for an account as of the reporting date the Marginal PD rate gets fixed ▸ This means that ratings transitions going forward are not assumed ▸ This method is very easy to implement but is simplistic in its assumption of constant Marginal PD rate ▸ $UnconditionalCumulativePD(YearN) = 1 - (1 - MarginalUnconditionalPD_{Yr1})^N$
Using Transitional Matrices	▸ A better alternative is to use the Rating Transition Matrix. ▸ Rating Transitions are natural in the Non-Retail world and therefore it makes it intuitive to assumed that rating will remain the same as at the reporting date but has a chance of moving upwards or downwards ▸ In this case the cumulative PD for any year N is derived by multiplying N annual transition matrices (since Transition Matrices are normally annual) Cumulative 3- Year PD for any Rating Grade (CUM_PD_3_YR) = (Uncond. Transition Matrix) ✖ (Uncond. Transition Matrix) ✖ (Uncond. Transition Matrix)

1. Possible Options for Unconditional PD Curves (Contd.)

Preferred Option for Unconditional PD Curves

- Given the 2 possible options described in the previous slide, the choice can be made based on the following criteria:
  - Rating Transition Matrix approach for Non-Retail accounts preferred to the maximum extent possible as it reflects the real dynamics of movement in credit risk of accounts on forward looking basis
  - In cases where we are not able to use the Rating Transition Matrix approach, we will use the No Rating Transition approach.
- Using the Rating Transition Matrix Approach obviously means that we need to have a TTC or Unconditional Rating Transition Matrix
- There are two possible ways to achieve this:
  - Derive the TTC Transition Matrix using Internal Rating Transition data
  - Use the S&P or Moody's Long Term Average transition matrix

1. A. Choice of the Transitional Matrix

Using Internal Ratings Data	▸ Deriving TTC Transition Matrix using internal rating transition would be a challenge given that for the Non-Retail portfolio data extensive data would not be available. ▸ In addition, if the internal rating system of the Bank has undergone changes in the recent past this approach wont be useful. ▸ Also even for say 7 rating grades we are talking about a 7*7 i.e. 49 cell transition matrix. Estimating each of the 49 cells using internal data would be extremely difficult
Using External Transition Matrices	▸ Perform a mapping of internal rating grades to external rating agency grades. ▸ S&P and Moody's given their large global datasets on rating transitions, provide Long Term Average transition matrices and also point-in-time matrices for different years ▸ These agency transition matrices can form the basis of rating transition models.

Conditioning the PD Curve

2. Conditioning PD based on Macro Factors

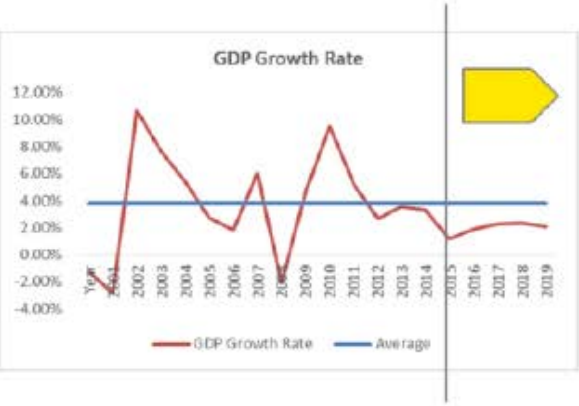
- In the previous section, we had described how to derive the Unconditional PD curve based on Rating Transition Matrices. In this section we talk about how to condition the Unconditional PD curve based on macro-factors
- The process of conditioning of PD curve based on macro factors is a 2 step process:
  - Determining which macro variables affect PD and their forward looking scenarios
  - Determining the Link Function between macro factor and PD
- The Link Function establishes the link between the Macro Factors and the Unconditional PD to give the Conditional PD. There are two possible and commonly used options for Link Functions:
  - Vasicek Framework
  - Regression Based Framework

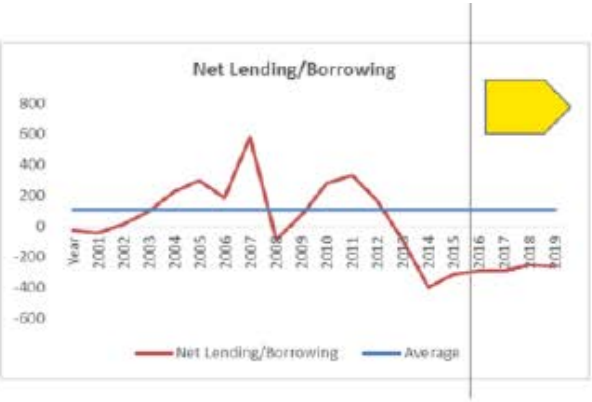
Preferred Option for Link Function

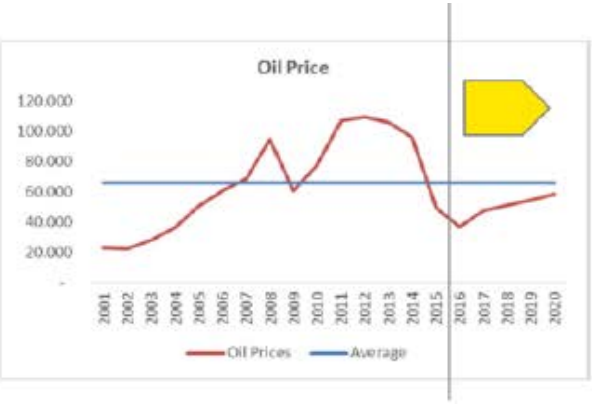
- Given that default data is sparse in case of Non-Retail portfolio to run robust regressions, Vasicek Framework would be more suited for Non-Retail and is commonly used.
- An example of how the conditioning is done is given in the next slide

Illustration of how Vasicek Adjustment is Done

- The Vasicek model, a popular framework that underpins the Basel IRB Economic Capital formula, is used for linking the Systemic Macroeconomic Factor with the Conditional Point-In-Time PD
- In this example we choose GDP Growth Rate, Net Lending/Borrowing and Oil Price as macro factors with equal weighting to each of them







Rating Grade	PD as per Credit policy
2	0.01%
3	0.05%
4	0.21%
5	0.86%
6	3.57%
7	14.90%

Portfolio Average 1.96%

Long Term TTC PD *Systemic Factor*

$$P_i = \emptyset\left[\frac{\emptyset^{-1}(P_\mu) - \sqrt{\rho}Z}{\sqrt{1-\rho}}\right]$$

Asset Correlation

Vasicek Model

Rating	PD Post Macro Adjustment
2	0.02%
3	0.10%
4	0.44%
5	1.75%
6	6.33%
7	22.58%

Portfolio Average 2.88%

Using the Link Function to derive Conditional PD

- We will adjust the Unconditional PD curve as follows:
  - A TTC PD will be established for the Unconditional PD curve based on the history of data used for the NTRS adjustment and Vintage Curve.
  - For example if data history of 2011 to 2015 is used for estimation of Unconditional PD curve and the chosen Macro factor is GDP Growth Rate, then the average GDP Growth Rate from 2011 to 2015 will be the TTC Baseline
  - Taking the above example forward, if the forecast GDP Growth Rate for year 3 was GDPyear3, then the Conditional PD for year 3 will be derived as follows:
  - $ConditionalPD_{yr3} = UnconditionalPD_{yr3} + \frac{\Delta PD}{\Delta GDPGrowth} * \left(GDP_{yr3} - TTCPD\right)$

PD Modelling

Questions to Consider

- Calculate multi-year PDs (including non-integer years, such 3.5 years) using the marginal PDs given in the marginal approach slide for any one segment
- Given certain coefficients and factor values, calculate the predicted output of a logistic regression model output, i.e., PD
- Calculate the Kaplan-Meier curve given a series of time v/s defaults data series

6 Forward Looking Adjustments

Forward Looking Adjustments

IFRS 9 Considerations

PiT Estimates

The PD values derived so far, apart from the fact that they can be used for lifetime calculation, are unconditional and TTC, a feature shared by Basel PD models. IFRS 9 requires PD estimates to be conditional on relevant, forward looking information

Non Macro

IFRS 9 only suggests macroeconomic forecasts as one potential source of relevant, forward looking information. Other information specific to the portfolios, a sub-pool of exposures or the bank could also be considered to obtain conditional PiT estimates

Overlays

Closely related to the previous consideration, the effect of the forward looking information can also be incorporated through overlays. IFRS 9 does not require the adjustment due to forward looking information as necessarily a 'model'

Weighted Scenarios

Given that multiple forward looking forecasts can potentially be provided, a probabilistic weighting scheme needs to be considered, thus obtaining a weighted average estimate

Forward Looking Adjustments

A Couple of Approaches

Macroeconomic Model 1

Leverage Stress Test Models / Approaches

- Macroeconomic data a must
- Robustness of model depends on richness of data
- Usually, a strong relationship for PD is obtainable
- A strong relationship for LGD may not always be possible

Establish a regression based relationship between PD values and potential macroeconomic factors over a TTC period of 5 – 7 years

Obtain forecasts for macroeconomic variables part of model

Adjust unconditional PD values based on output from regression model output; calibrate adjustment based on TTC levels

Judgment Overlay 2

Effort Intensive, but more flexible

- Requires careful thought, consideration and usually, expert involvement
- Output more flexible than regression based equation
- Can be updated easily, quickly and without considerable cost or effort
- Will most likely be needed for LGD adjustments

Probabilistic Weighted Scenarios

An 'Expected' Approach to ECL Estimation

Considerations

- ECL, LECL calculation should be separately performed under alternative macroeconomic scenarios to reflect a range of potential future outcomes
- Probability weights should be applied to the different scenarios for the calculation of the final, single, consolidated ECL
- Consideration of PD values, or otherwise, for staging rules also need to take probabilistic weighting into account
- For a single account, the weighted ECL is 12m or Lifetime based on the staging criteria

Weights reflect a normal distribution of probability

	Severe Stress	Mild Stress	Baseline	Mild Optimism	Severe Optimism
Weightage	5%	20%	50%	20%	5%
12m PD	5%	4%	3%	2%	1%
Lifetime PD	18%	15%	10%	6%	4%
LGD	70%	60%	50%	40%	30%
EAD	1,000	1,000	1,000	1,000	1,000
12m ECL					
LECL					
Weighted 12m ECL	16				
Weighted LECL	55				

Forward Looking Adjustments

Questions to Consider

- Calculate scenario weighted 12m and Lifetime ECL estimates given the information in the table. Assume constant marginal PD approach is used

7 LGD Modelling

Loss Given Default

IFRS 9 Considerations

Downturn
IFRS 9 removes the requirement to calculate / calibrate a 'downturn' LGD, as was the case under Basel requirements

Discount
For discounting of cash flows, IFRS 9 requires the usage of Effective Interest Rate as opposed to other alternatives that could possibly used under Basel, such as WACC

Costs
Basel required computation of LGA to be 'economic', i.e., consideration off costs, including indirect. IFRS 9 requirements deem non-consideration of indirect / general costs

PiT
Just like the PD estimates, LGD estimates also need to PiT and thus, take into account relevant forward looking information

General Viewpoints

- As can be seen from the considerations, LGD modelling doesn't require as much deliberation as PD
- For the most part, Basel approaches, if not models, can be leveraged greatly, with appropriate adjustments based on required considerations
- The PiT adjustment through forward looking information can be challenging. Based on observed practices, obtaining strong relationships between LGD and macro economic data can be difficult
- As such, a more judgmental overlay approach (as outlined in a previous slide) will likely be applied